

Candidates' Performance

Paper 1A

This section consisted of 40 multiple-choice questions. Overall, candidates performed satisfactorily, with an average of 28 questions answered correctly. Their performance was notably stronger in 'Computational Thinking and Programming,' while they struggled more in 'Internet and its Applications.' A post-examination item analysis revealed the following insights:

1. Question 9 aimed to assess candidates' understanding of common file formats, specifically PDF and HTML. Half of the candidates were able to demonstrate an understanding of the advantages of PDF over HTML. However, some candidates selected options that clearly favored the HTML format in applications, indicating a lack of comprehension regarding the significance of HTML in web contexts.

Q.9 John publishes an article on the Internet. He considers using PDF instead of HTML because

- A. a browser can be used to open the article. (9%)
- B. multimedia elements can be added to the article. (19%)
- C. the file size is smaller. (20%)
- * D. it offers better document protection. (52%)

2. Question 12 aimed to assess candidates' knowledge of operating system functions. Fewer than half of the candidates answered correctly, though approximately two thirds of the candidates recognised that system security is a core function of an operating system.

Q.12 Which of the following are the functions of an operating system?

- (1) Managing input and output devices.
- (2) Managing system security.
- (3) Managing computer's memory.
- A. (1) and (2) only (10%)
- B. (1) and (3) only (36%)
- C. (2) and (3) only (9%)
- * D. (1), (2) and (3) (45%)

3. Question 13 aimed to assess candidates' understanding of virtualisation. Half of the candidates answered correctly, while others confused virtualisation with storage aggregation, suggesting a lack of clarity in differentiating between virtual machines and RAID-like systems.

Q.13 Which of the following are examples of applications of virtualisation?

- (1) Creating a virtual environment using a different operating system within the current operating system.
- (2) Partitioning a physical server into several virtual servers for running multiple application programs.
- (3) Combining several physical hard disks into a single virtual storage device.
- * A. (1) and (2) only (52%)
- B. (1) and (3) only (8%)
- C. (2) and (3) only (16%)
- D. (1), (2) and (3) (24%)

4. Question 18 aimed to assess candidates' basic understanding of IPv4 addresses. Fewer than half of the candidates were able to identify that each number in an IP address is an integer ranging from 0 to 255, a fundamental aspect of network management. This concept, alongside the distinction between the network and host portions, is crucial for grasping the IPv4 structure. The results showed that the majority of the candidates did not fully understand the representation of IP addresses.

Q.18 Which of the following is/are valid IPv4 address(es)?

- (1) 10.512.10.0
- (2) 1.2.3.4
- (3) 192.168.1.256

- A. (1) only (6%)
- * B. (2) only (44%)
- C. (1) and (3) only (23%)
- D. (2) and (3) only (27%)

5. Question 24 aimed to assess candidates' ability to integrate knowledge of TCP/IP applications. Half of the candidates understood IP routing. The other candidates erroneously assumed that IP routing involved fixed paths, which relates to circuit switching rather than packet switching.

Q.24 Which of the following statements about transmitting data using TCP/IP are correct?

- (1) An IP address is used to identify a device in a network.
- (2) Data packets are reassembled back to the original data at the destination.
- (3) The IP determines a fixed path to send data packets to the specific destination.

- * A. (1) and (2) only (49%)
- B. (1) and (3) only (9%)
- C. (2) and (3) only (16%)
- D. (1), (2) and (3) (26%)

6. Question 38 aimed to assess candidates' ability to use conditional loops to solve problems. 65% of candidates answered the question correctly. The remaining candidates might trace the conditional increments incorrectly, particularly in odd/even handling.

Q.38 What is the output of the following algorithm?

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S ← 0
A ← 3
while A ≤ 9
    S ← S + A
    if (the remainder of (A/2)) = 1
        A ← A + 1
    else
        A ← 2 * A + 1
output S

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- A. 7 (14%)
- B. 9 (5%)
- C. 10 (16%)
- * D. 16 (65%)

Paper 1B

Question Number	Performance in General
1 (a)	Satisfactory. Around half of the candidates applied the correct formula with accurate cell references. A common error was using 'AVG' instead of the appropriate function name.
(b)	Fair. Candidates struggled with concatenation, particularly in using the '&' operator to combine comparison operators with cell addresses in the parameters.
2 (a)	Satisfactory. While candidates accurately described what a one-time password (OTP) is, only a few demonstrated familiarity with its key characteristics.
(b)	Satisfactory. Some responses omitted specifying who performs the encryption/decryption or who owns the cryptographic key.
3 (a)	Fair. Frequent errors included failing to convert bytes to bits (1 Byte = 8 bits) or neglecting to include the unit (seconds).
(b)	Fair. In (i), candidates in general identified that the two URLs were the same but did not elaborate further or discuss the role of IP addresses. In (ii), candidates were unclear about the operational differences between downloading a file via a link and an email attachment.
4 (a)	Poor. In (i), candidates demonstrated a limited understanding of HTML code application, particularly in distinguishing between display size and file size. In (ii), only a small number of candidates accurately described the impact of changes to the photo's aspect ratio.
(b)	Poor. Responses indicated a narrow grasp of AI concepts. Candidates were unfamiliar with key machine learning considerations (e.g. selecting photo samples for training) and often focused on irrelevant factors like copyright or data privacy.
5 (a)	Satisfactory. Weaker candidates occasionally misused technical terms, but most responses were acceptable.
(b)	Fair. Only a small number of candidates identified the core function of the driver program.
6 (a)	Very good.
(b)	Good.

Question Number	Performance in General
7	(a) Satisfactory. Weaker candidates did not use proper terms for form elements, and a few misunderstood the role of dropdown menus, incorrectly stating that they ensure input correctness rather than input reasonableness. (b) Satisfactory. In (i), candidates provided multiple answers from the same category, while weaker responses were overly simplistic without elaboration. In (ii), weaker candidates used imprecise terms like 'system' instead of 'operating system.' (c) Poor. Only a small number of candidates correctly identified 'fixed value check,' and some mistakenly believed that multiple validation checks could apply to a single data value. (d) Very good.
8	(a) Satisfactory. In (i), weaker candidates confused array content with indices, leading to incorrect values for Smax and Smin. In (ii), they inaccurately described the data type as 'numeric' without justification and omitted explanations entirely. (b) Satisfactory. In (ii), weaker candidates gave vague answers like 'easy to change' without elaboration. (c) Fair. Candidates struggled with program segments, featuring incorrect loop ranges (e.g. range(2, N) instead of range(2, N+1)), syntax errors, or uninitialised variables (e.g. Smax, Smin and FS).
9	(a) Good. (b) Fair. In (i), weaker candidates demonstrated unfamiliarity with check digit concepts, with many failing to justify their valid answers. In (ii), they were unable to provide relevant examples for the scenario. (c) Fair. Approximately two thirds of the candidates misunderstood the application of flags in this context, and only one third successfully identified all errors in the given problem. (d) Fair. Weaker candidates struggled in particular, often incorrectly discussing features of freeware and shareware instead of addressing the actual reasons for giving up on the software installation.

Paper 2

Section A Databases

Question Number	Performance in General
1 (a)	Good. Candidates were able to complete the ER diagram correctly, although weaker candidates were unable to provide the minimal cardinalities.
(b)	Fair. Candidates were unfamiliar with the functions of rollback in a DBMS. Weaker candidates failed to articulate any benefit of rollback.
2 (a)	Poor. Nearly all candidates were unable to use the UNION command in SQL. Additionally, weaker candidates did not understand that the UNION command selects only distinct values by default.
(b)	Good. Candidates successfully updated the value of AMT for a specific record in T1.
3 (a)	Fair. A high proportion of candidates incorrectly used the INSERT INTO command, often misapplying the proper quotation marks for data values.
(b)	Satisfactory. The majority of candidates provided the correct SQL statement. However, weaker candidates failed to apply the SUM function properly.
4	Fair. Only a small number of candidates successfully completed the two normalised database tables with appropriate attributes and provided the correct primary keys.
5 (a)	Satisfactory. Candidates generally assigned the correct data types, citing 'leading zero' as a reason.
(b)	Fair. In (ii), weaker candidates failed to provide a proper predicate. In (iii), more than half of the candidates encountered difficulties with subqueries and the MAX aggregate function.
(c)	Satisfactory. Candidates generally used the SQL statement correctly, although weaker candidates confused MINUS and MIN.
(d)	Good. A high proportion of candidates clearly identified and described the integrity problem clearly.
(e)	Fair. A small number of candidates accurately described the process of denormalising two database tables and its related benefits. Weaker candidates confused the concepts of decomposing tables and denormalising tables.

Section B
Web Application Development

Question Number	Performance in General
6 (a) (b)	Satisfactory. More than half of the candidates were able to effectively apply three different types of input controls to meet the search criteria requirements. However, a few candidates failed to suggest user-friendly input designs or did not recognise that the input for spending per head should be a range. Poor. Fewer than one third of the candidates were able to give correct reasons for using external CSS. The results indicated that over two thirds of the candidates lacked adequate knowledge of CSS, and only a very small number of candidates understood the use of inline CSS well.
7 (a) (b)	Satisfactory. Around half of the candidates demonstrated familiarity with folder management and the access rights related to directories, folders, and files. However, only 20% of the candidates suggested solutions to allow individuals to view only their own photos stored in a folder. More than half of the candidates incorrectly suggested adding a password to the photos. Poor. Only a small number of candidates were able to give the pros and cons of the two methods for sharing a printer in a network. Slightly more than 10% of the candidates recognised that Method 2 allows for greater control over the printer.
8 (a) (b)	Fair. Fewer than one third of the candidates were able to identify the functions of certain network devices and provide correct functions for the two devices, A1 and A2. Weaker candidates incorrectly provided names instead of functions. Poor. Only a small number of candidates were able to accurately identify the names of the devices and their corresponding functions. More than half of the candidates listed only the name without identifying their functions, with weaker candidates incorrectly suggesting router, modem and AP.

Question Number	Performance in General
9	(a) Satisfactory. Around half of the candidates were familiar with setting the background colour and font size of a web page. However, only one quarter of the candidates demonstrated experience using the 'onload' event to execute a script after a web page fully loads. (b) Poor. Only a small number of candidates correctly identified the missing part. More than two thirds of the candidates were unaware of the characteristics of ASCII code and failed to use it to assign a value to the temporary variable. (c) Poor. Only approximately one quarter of the candidates were familiar with SQL server connections and the code to check those connections. More than half of the candidates did not recognise that Line 204 would produce an error message, preventing the remaining scripts from running if the connection to the SQL server (database) fails. (d) Poor. Only approximately one quarter of the candidates were able to use the 'order by' command. (e) Fair. Around half of the candidates used '\$row[' ITEMS ']== ' or 'Strcmp' to compare two strings to identify the rows that needed to be highlighted in grey. (f) Poor. Only a small number of candidates demonstrated experience with the function 'mysqli_num_rows()', while weaker candidates incorrectly used 'count' as their answer. (g) Very poor. Only a very small number of candidates demonstrated experience with the function 'mysqli_fetch_array()' and aggregating values of variables upon certain conditions. (h) Very poor. Only a very small number of candidates showed familiarity with the function 'mysqli_close()' to prevent improper closing of the connections, which can lead to resource leaks, connection limits, performance issues, data integrity risks, and security vulnerabilities.